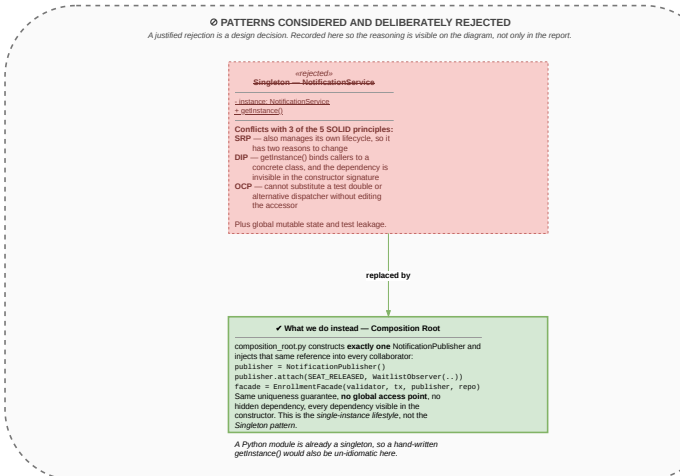
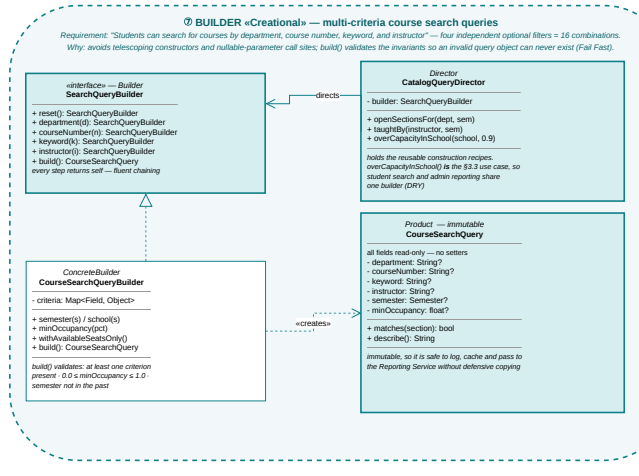
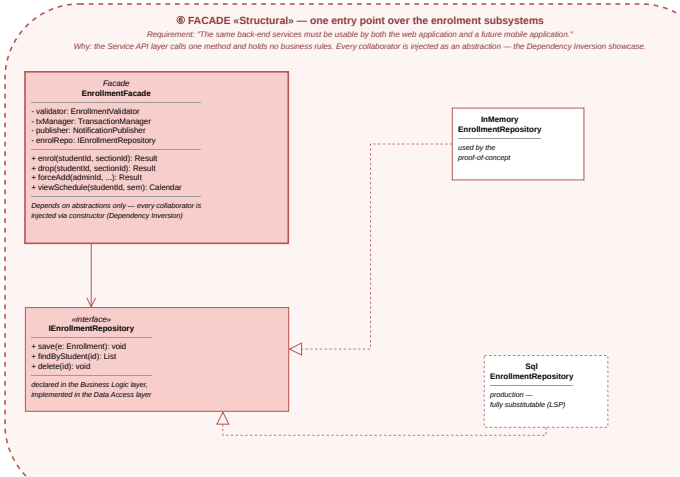
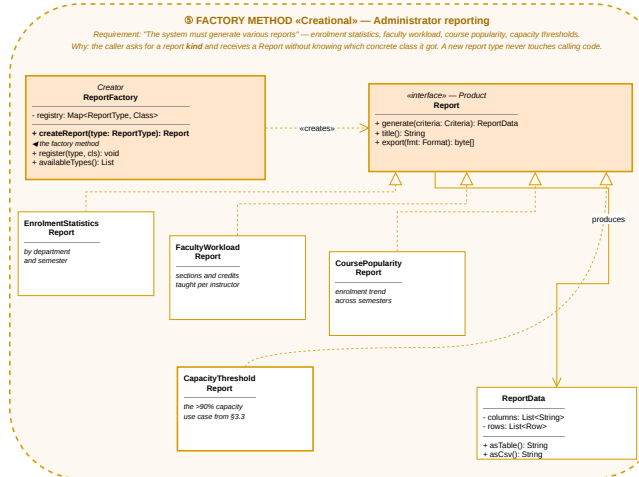
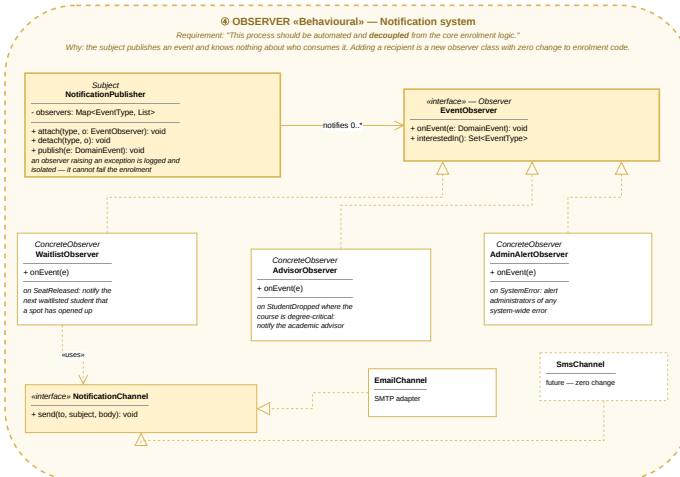
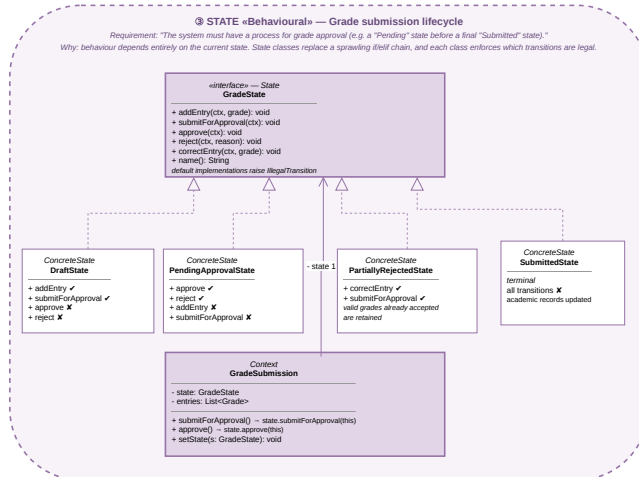
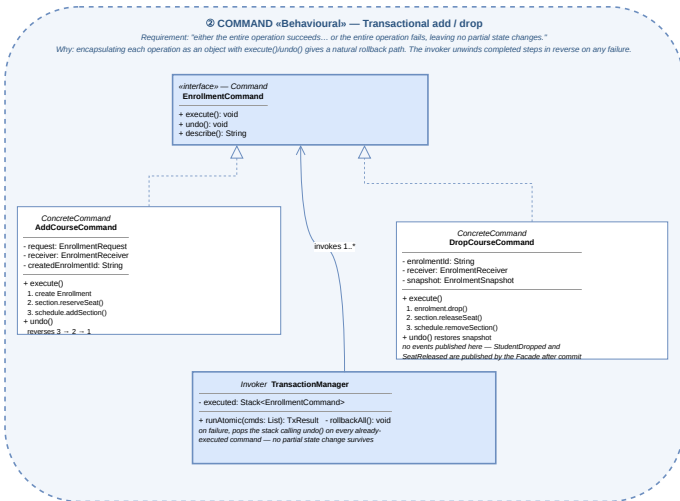
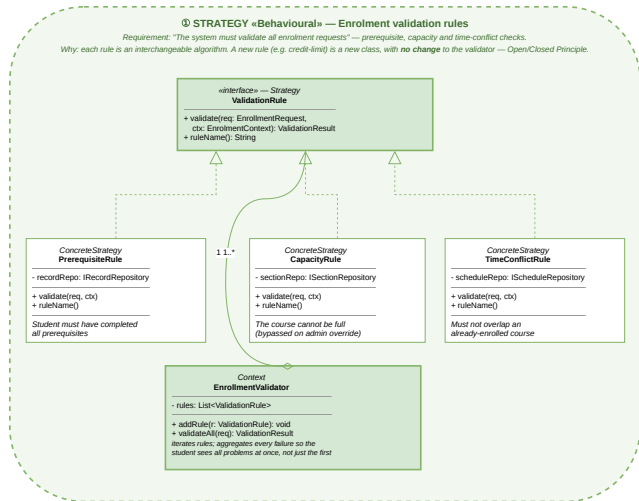


NexusEnroll — Design Pattern Structures in the Business Logic Tier

Seven patterns across all three GoF families. Each zone names the pattern, its GoF roles, and the specific requirement it satisfies. The final panel records the patterns we considered and rejected, and why.



Client-side call — the \$1 use case
"Browse all computer science courses for the upcoming semester that a specific professor teaches"

```

query = (CourseSearchQuery.builder()
    .department("Computer Science")
    .instructor("Prof. Silva")
    .semester(Semester.FALL_2026)
    .build())

```

vs. without Builder
search(browse, None, None, "Prof. Silva", FALL_2026, None, False)
— a transposed argument is a silent bug, not an error

Also evaluated and rejected — full reasoning in \$6 of OI design patterns and
Abstract Factory one product family only, so Factory Method is the right granularity (YAGNI), Decorator for chaining validation rules — a decorator chain short-circuits, but the requirement benefits from reporting every failure
Proxy may violate locality — no requirement drives it, premature optimisation (KISS), Template Method report skeleton — closest call, the four reports offer in aggregation shape, not in one substitutable step, Chain of Responsibility approval escalation — the university has single-administrator approval, not a hierarchy

How the patterns collaborate at run time (see diagram 05 — Sequence)

EnrolmentFacade ⑥ → asks EnrolmentValidator ③ to run every ValidationRule ⑤ → wraps the state changes in an AddCourseCommand ② handed to TransactionManager ②
on commit, publishes a domain event through NotificationPublisher ⑧ — each EventObserver ⑦ reacts independently.
GradeSubmission ③ moves Draft → PendingApproval → Submitted, delegating every transition to its current GradeState ③, ReportFactory ⑤ and CourseSearchQueryBuilder ④ are invoked independently by the Administrator and Student modules.